Design Detectives: DataLab

PS50008A Week 12 Lab

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Welcome

Your Data, Your Discoveries

Lab Overview

Today we work through your **real DataLab results** as a class.

Structure

1. Task overview (me)
2. Small group discussion (you)
3. Whole-class debrief (together)
4. Repeat for each task

Your Job

- Form groups of 2-3
- Open the activity page
- Discuss the questions
- Be ready to share answers

Timeline

Phase	Task	Duration
1	Reflex Test	12 min
2	Time Warp	10 min
3	Memory	15 min
4	Frontal Lob	12 min
5	Sleep & Memory	12 min
6	Sixth Sense	10 min
7	Wrap-up	10 min

Total: ~80 minutes

Open the Activity Page

Navigate to **Week 12 > Lab > Activity**

You’ll see questions for each task.

Work through them with your group.

Phase 1: Reflex Test

How fast are your reactions?

What You Did

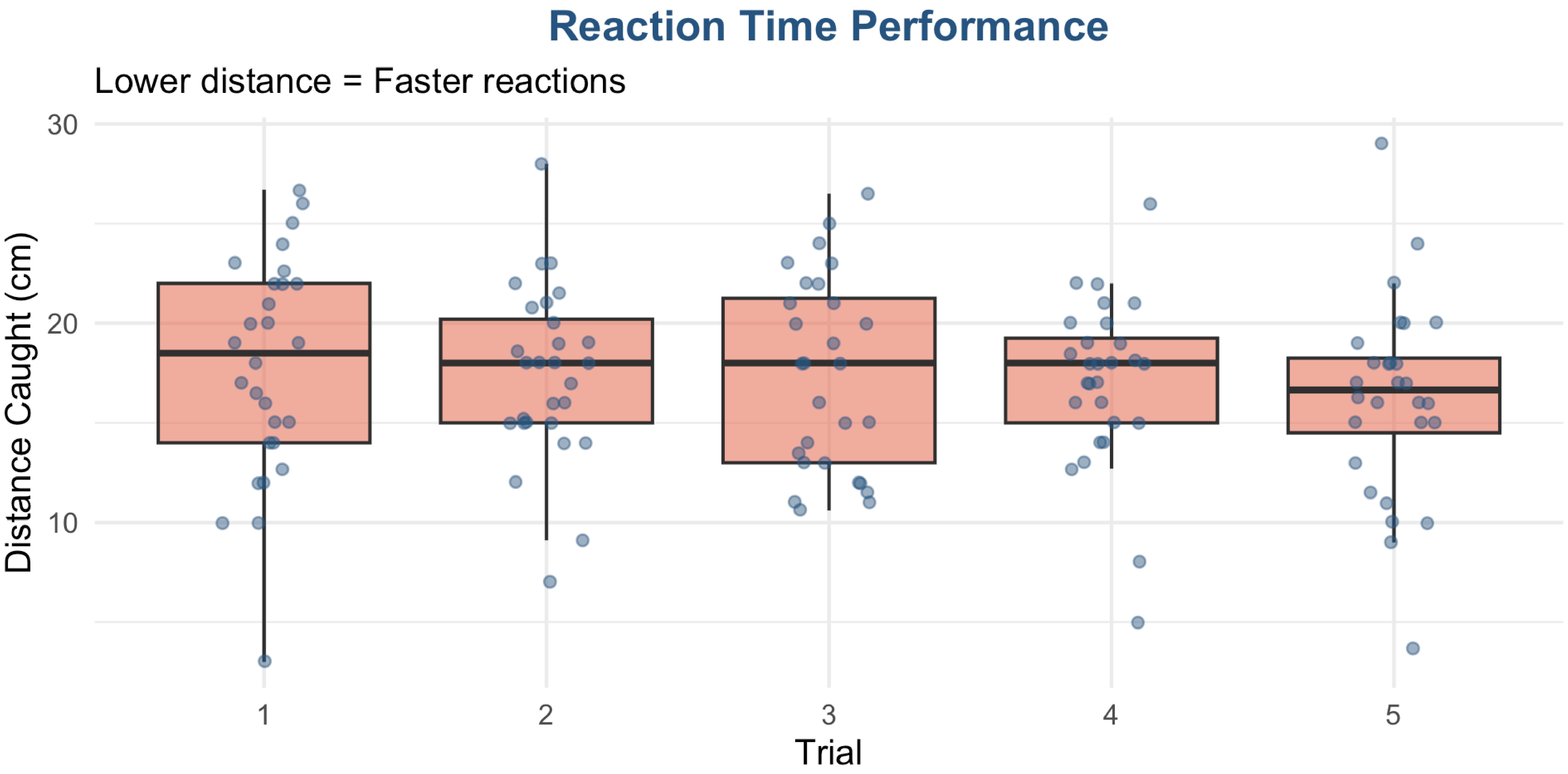
You caught a falling **30cm ruler** five times.

The distance caught reflects **reaction time**:

$$\text{RT (ms)} = \sqrt{\frac{2 \times \text{distance (cm)}}{980}} \times 1000$$

Shorter distance = faster reactions

Your Results



Group Discussion

In your groups, answer Questions 1.1-1.6:

1. What pattern do you see from Trial 1 to Trial 5?
2. What design type is this? (Between/Within/Correlational)
3. What is the IV? What is the DV?
4. Write a hypothesis for the learning effect.

04:59

Debrief: The Answers

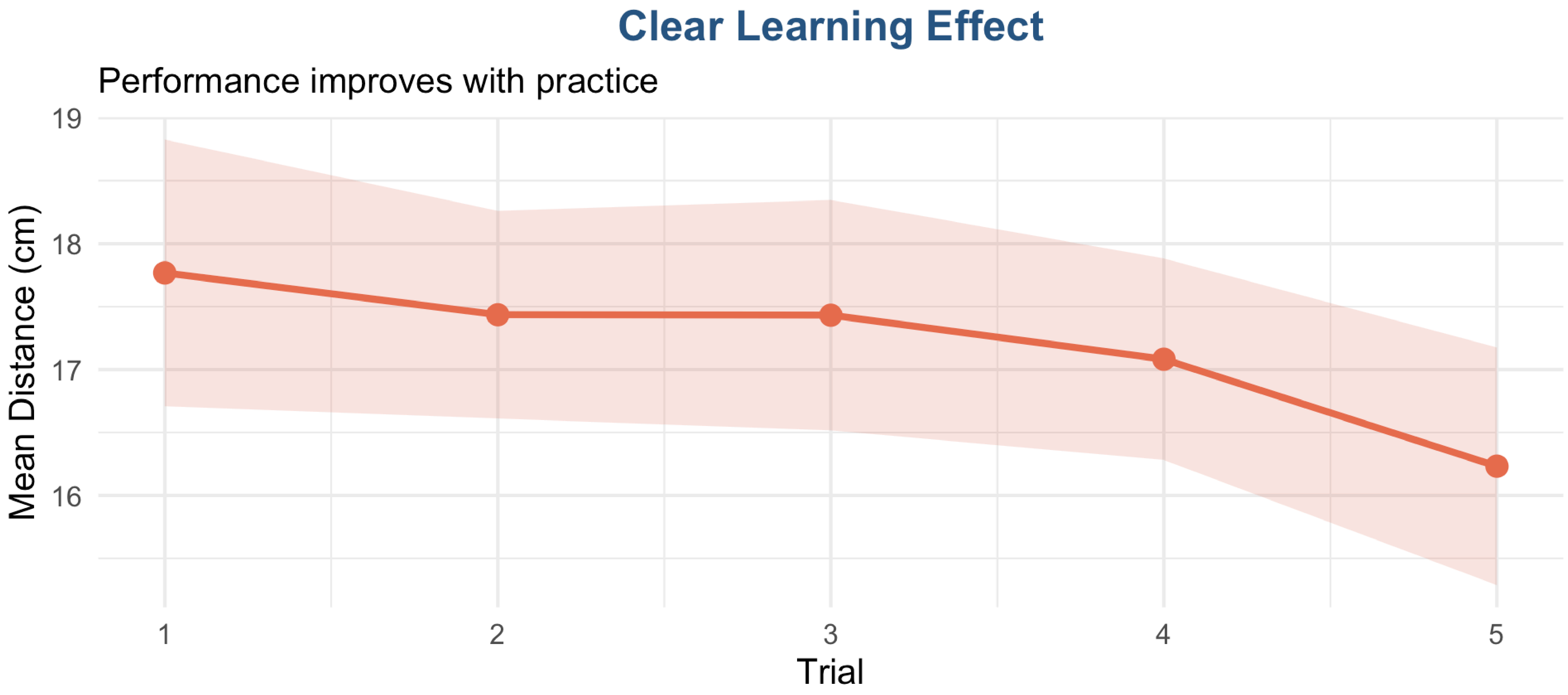
Design: Within-subjects (same people, all trials)

IV: Trial number (1, 2, 3, 4, 5)

DV: Catch distance (cm) or reaction time (ms)

Hypothesis (one-tailed): Reaction times will decrease from Trial 1 to Trial 5.

The Learning Effect



Why does this matter? Practice effects are a confound we control for in experiments.

Phase 2: Time Warp

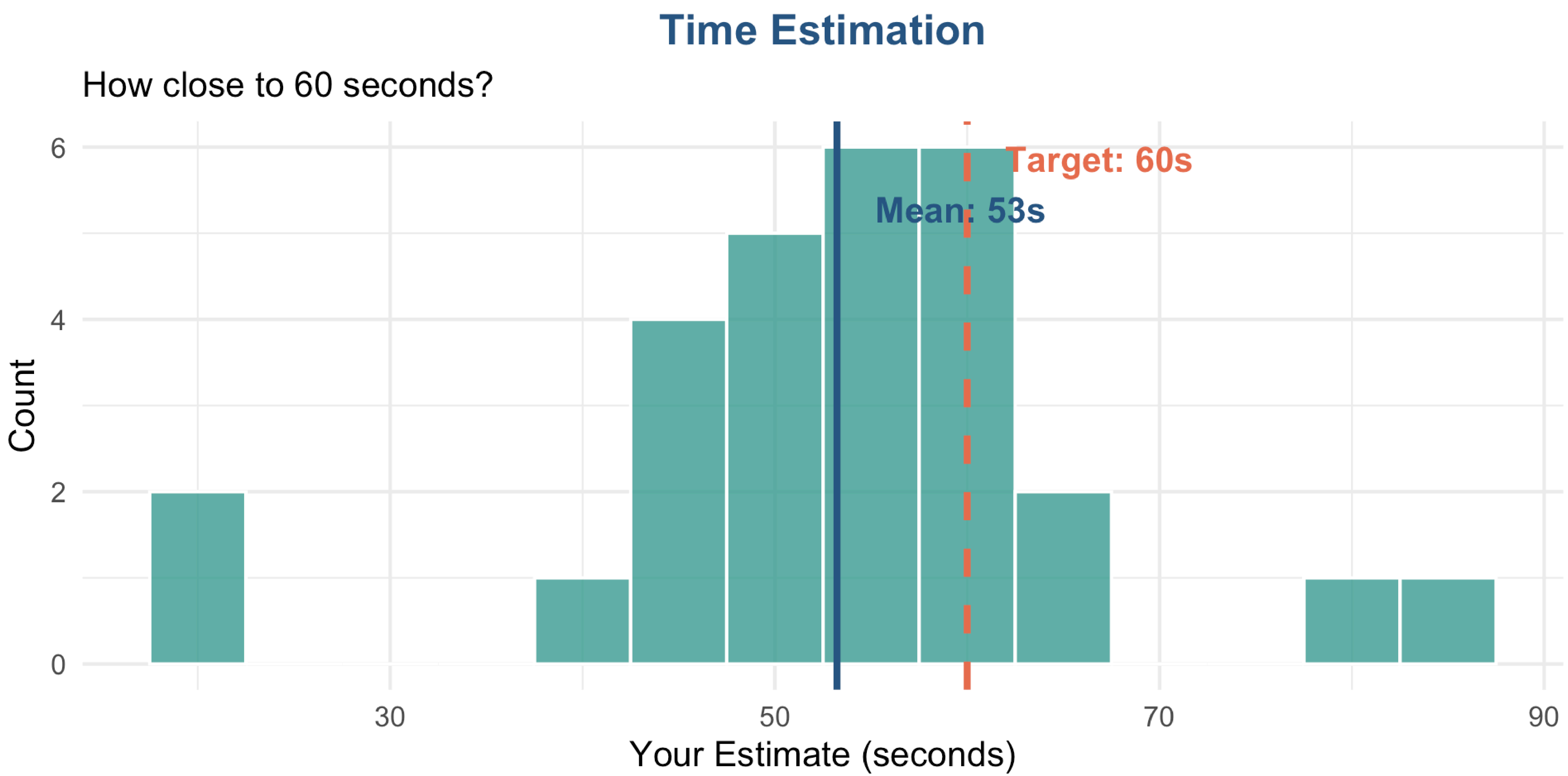
How well do you perceive time?

What You Did

You estimated **60 seconds** without counting.

Then you estimated the length of lines and strings.

Your Time Estimates



Group Discussion

In your groups, answer Questions 2.1-2.3:

1. Did people over- or underestimate 60 seconds on average?
2. What design is this? (Between/Within/Correlational)
3. What would the null hypothesis be?

04:59

Debrief: Time Perception

Finding: Most people **underestimate** 60 seconds (say it's up before it actually is).

Why? When engaged in a task, time seems to pass faster.

Design: This is testing against a known value (60 seconds) - a one-sample design.

Null hypothesis: Mean estimate will equal 60 seconds (no systematic bias).

Phase 3: Memory

MindPalace

What You Did

You memorised **10 words** and tried to recall them.

But here's what you didn't know...

The Secret Manipulation

Lab A memorised:

ABSTRACT words

- Justice
- Freedom
- Wisdom
- Courage
- Truth

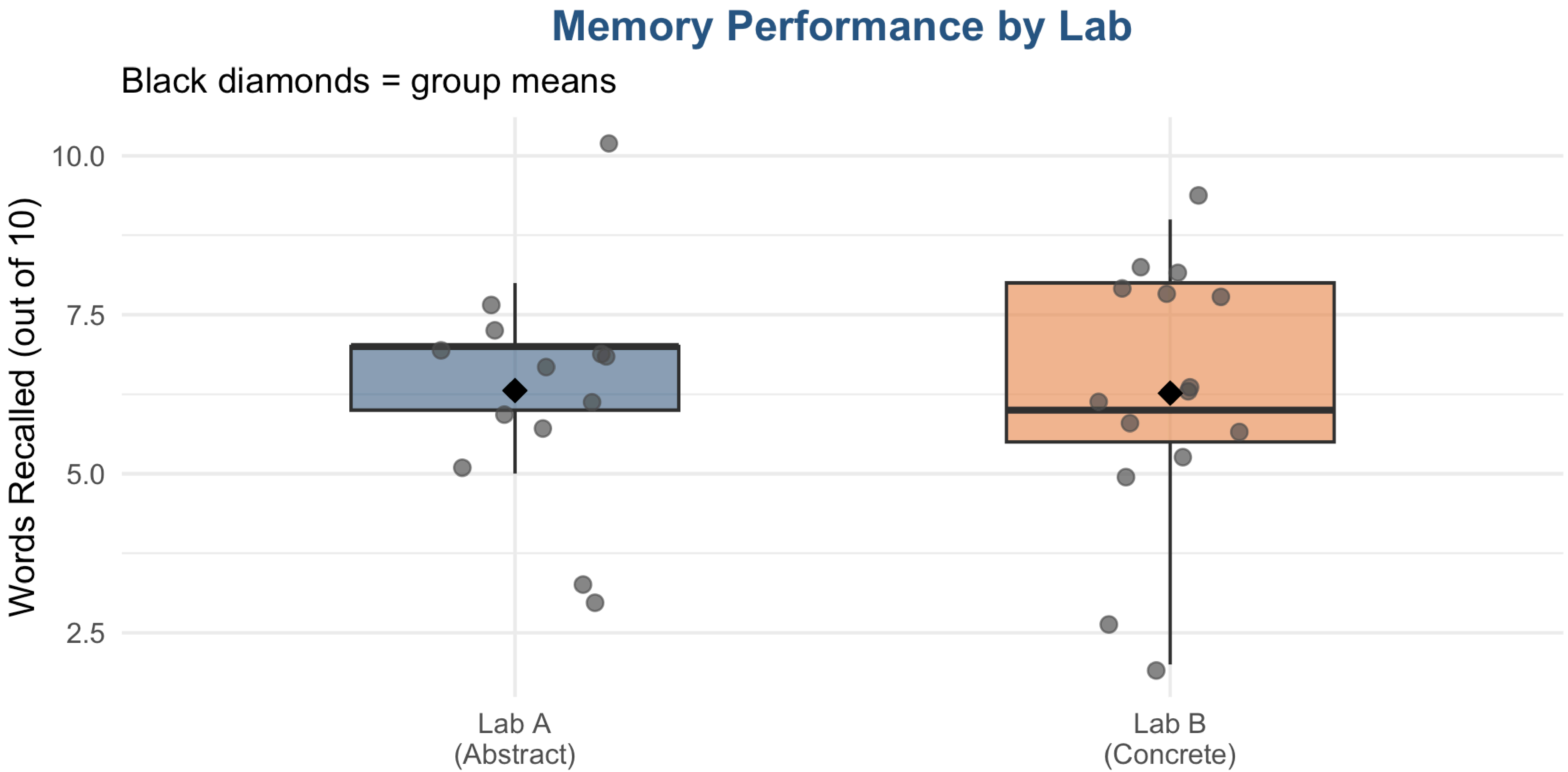
Lab B memorised:

CONCRETE words

- Chair
- River
- Mountain
- Apple
- Hammer

Question: Does word type affect memory?

Your Results: Lab A vs Lab B



The Statistics

Lab	n	Mean	SD
Lab A			
(Abstract)	13	6.31	1.89
Lab B			
(Concrete)	15	6.27	1.98

Group Discussion

In your groups, answer Questions 3.1-3.7:

1. What is the IV? What is the DV?
2. What design type is this?
3. Write an experimental hypothesis (one-tailed).
4. Write a null hypothesis.
5. Why might concrete words be easier to remember?

05:59

Debrief: Memory Design

IV: Word type (abstract vs concrete)

DV: Number of words recalled (0-10)

Design: Between-subjects (different people in each lab)

Hypotheses:

- **H1 (one-tailed)**: Concrete words will be recalled better than abstract words.
- **H0**: There will be no difference in recall between word types.

Why Concrete Words?

Dual Coding Theory (Paivio, 1971)

Concrete words can be stored TWO ways:

1. **Verbal code** (the word itself)
2. **Visual code** (a mental image)

Abstract words only have the verbal code.

Two codes > One code = Better memory

Phase 4: Frontal Lob

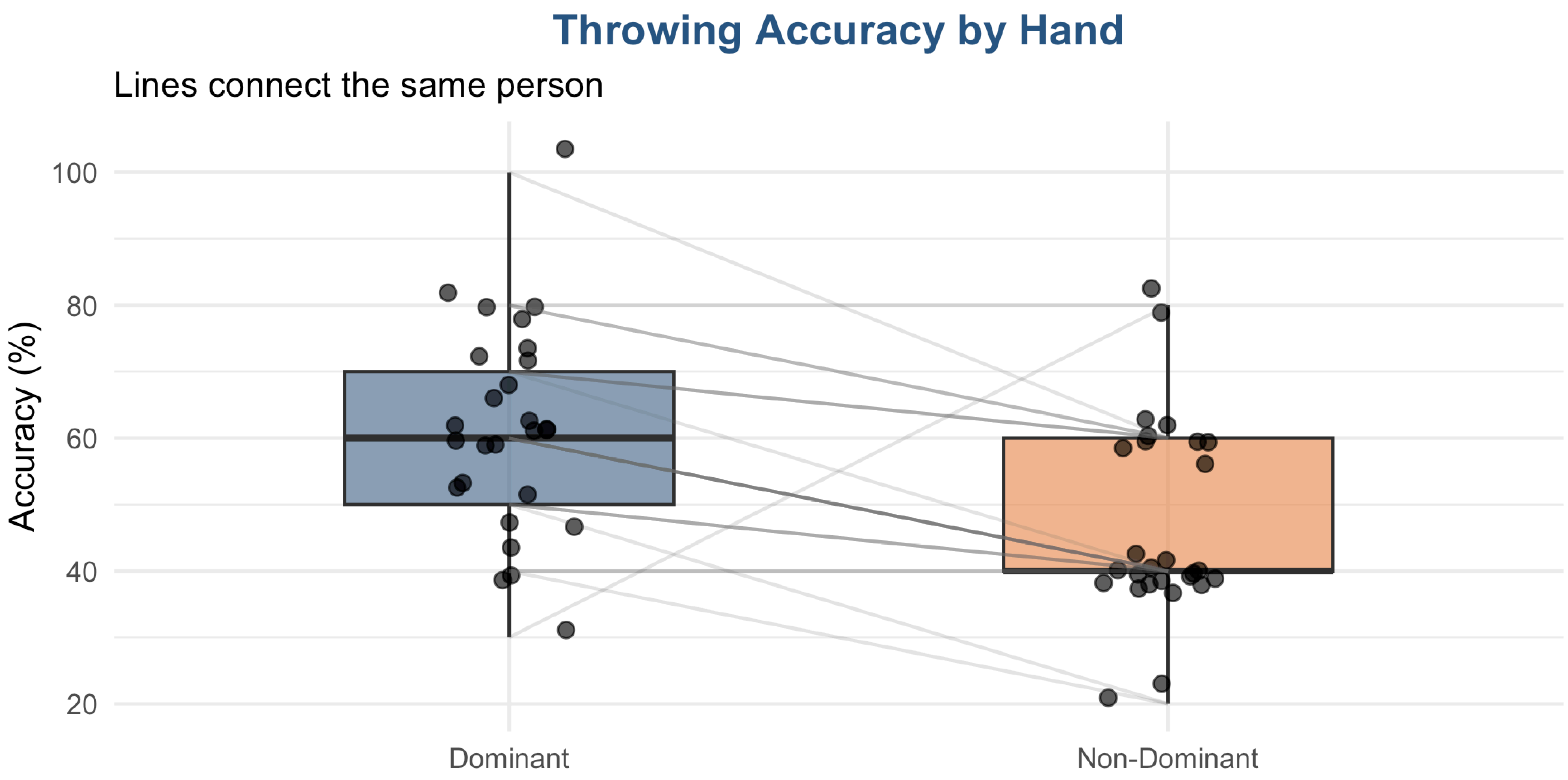
Does handedness matter?

What You Did

You threw balls at a bucket:

- **10 throws** with dominant hand
- **5 throws** with non-dominant hand

Your Throwing Results



Group Discussion

In your groups, answer Questions 4.1-4.6:

1. Same people or different people in each condition?
2. What design type is this?
3. Why is within-subjects “more powerful”?
4. What is the IV? What is the DV?

04:59

Debrief: Within-Subjects Power

Design: Within-subjects (same person, both hands)

IV: Hand used (dominant vs non-dominant)

DV: Throwing accuracy (%)

Why more powerful?

Each person is their own control. Individual differences (height, coordination, practice) are controlled because we compare YOU to YOU.

Phase 5: Sleep & Memory

Is there a relationship?

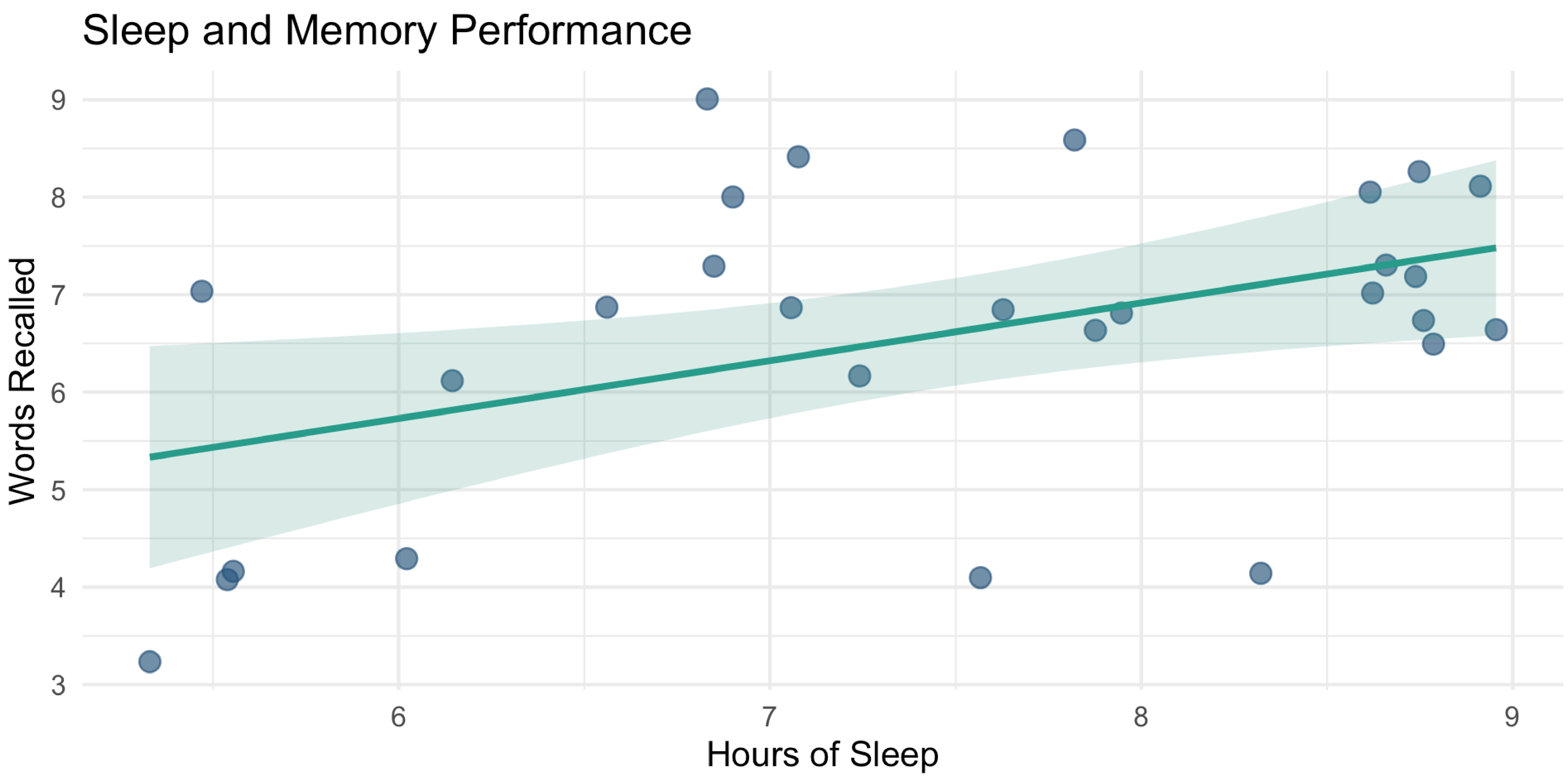
The Question

You told us how much sleep you got last night.

We measured how many words you recalled.

Is there a relationship between sleep and memory?

Your Results



Group Discussion

In your groups, answer Questions 5.1-5.5:

1. Is there an IV in this study?
2. What design type is this?
3. What does the correlation tell us?
4. Can we say sleep **causes** better memory?
5. What's the key limitation of correlational designs?

04:59

Debrief: Correlational Design

Design: Correlational (no manipulation)

Variables: Sleep hours AND memory score (neither is the IV)

Key insight: We measured BOTH variables — we didn't manipulate either.

⚠ **The Critical Point**

Correlation ≠ Causation

Maybe sleep helps memory… OR maybe third variables (e.g., health, motivation) affect both!

Phase 6: Sixth Sense

Can you feel being stared at?

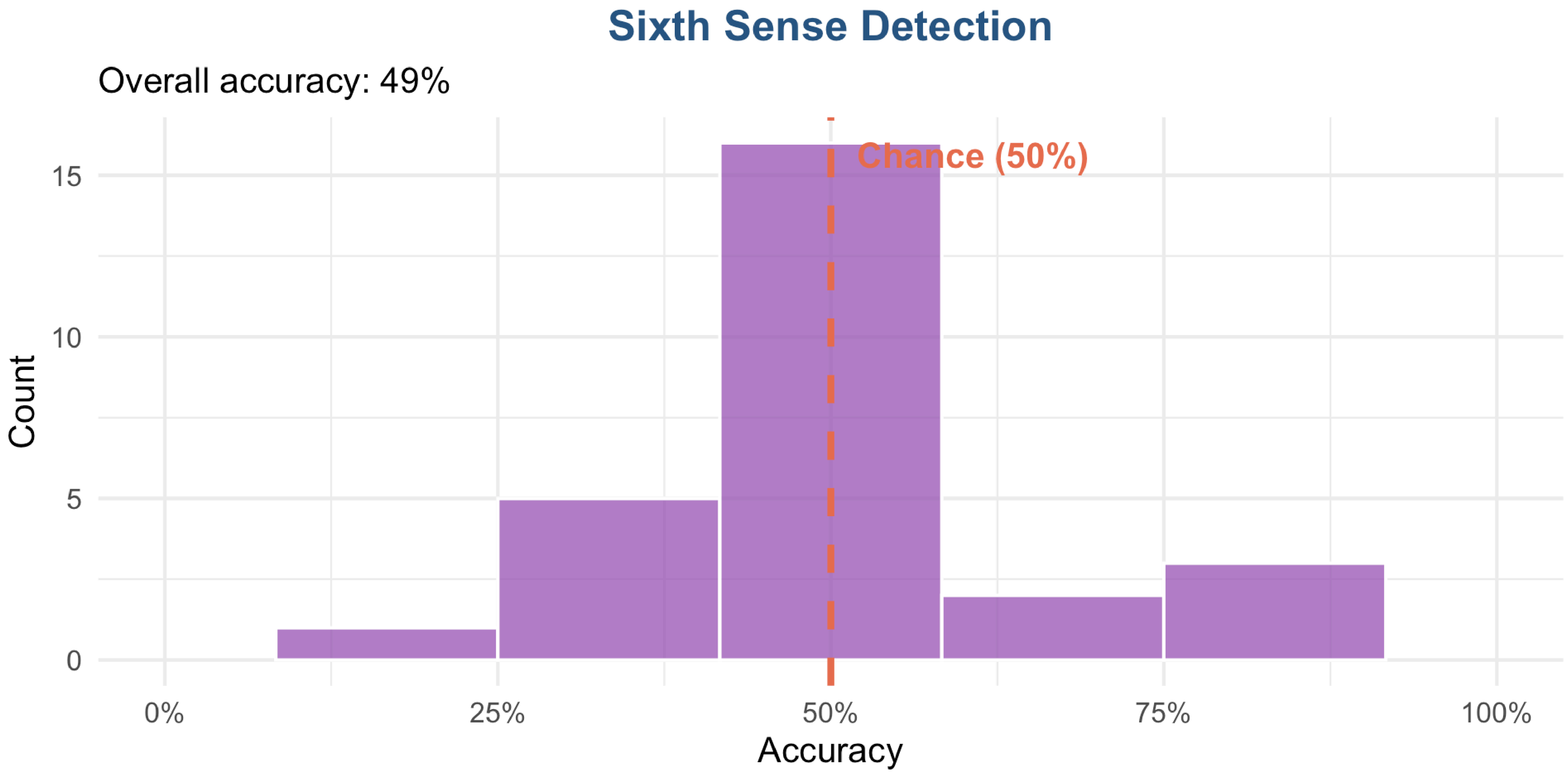
What You Did

6 trials: Someone either stared at you or didn't.

You guessed which it was.

If just guessing: You'd expect 50% correct (3 out of 6).

Your Detection Accuracy



Group Discussion

In your groups, answer Questions 6.1-6.4:

1. If someone got 5/6 correct, does that prove they have a sixth sense?
2. What is the null hypothesis here?
3. What would 50% accuracy suggest?

04:59

Debrief: Chance and the Null Hypothesis

H0: Performance equals chance (50% - just guessing)

H1: Performance differs from chance

Key insight: Some people will get 5/6 or 6/6 just by luck, even with no real ability.

With 28 people flipping coins, SOMEONE will get 5+ heads. That doesn’t mean the coin is biased.

This is why we need statistics - to separate signal from noise.

Wrap-Up

What have we learned?

Design Types Summary

Design	Same people?	Example from today
Between-subjects	No	Memory (Lab A vs Lab B)
Within-subjects	Yes	Throwing (Dom vs Non-dom)
Correlational	N/A	Sleep & Memory

Key Vocabulary

Variables

- **IV**: What's manipulated
- **DV**: What's measured

Hypotheses

- **H1**: Predicts an effect
- **H0**: Predicts no effect

Direction

- **One-tailed**: Predicts direction
- **Two-tailed**: Predicts difference only

Key Concept

- **Chance**: Random variation
- **Statistics**: Separating signal from noise

Your Assessments

Everything today connects to your assessments:

Assessment	What you practiced today
Research Plan	IV, DV, design types, hypotheses
MCQ Exam	All terminology and concepts
EDS Assignment	Means, SDs, interpreting results

Next Steps

1. Complete the **Week 12 Self-Assessment Quiz**
2. Review today's activity page in your own time and look up terms in the glossary
3. Read about the **Research Plan** (due Week 16)

Questions?

Well done!